Human Robot Teamwork Requirements

Structure of meeting

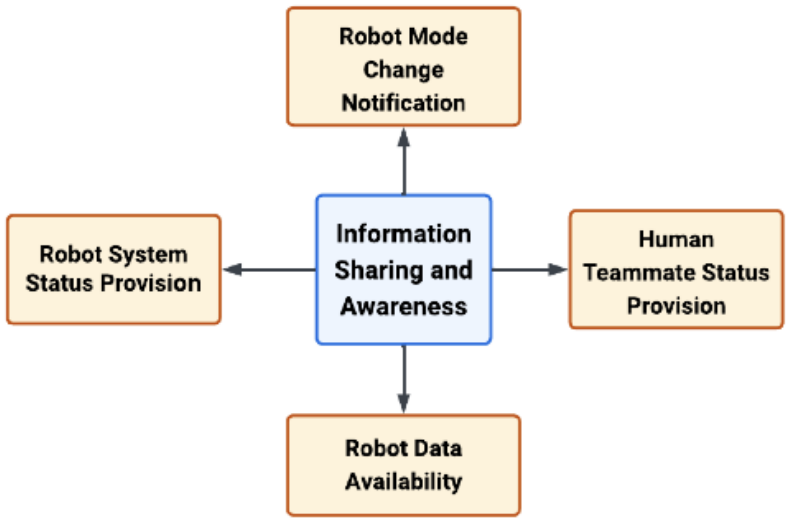
- Brief introduction to our classification of requirements
- Discussion of each of the 6 categories in turn
- Some general questions on the classification

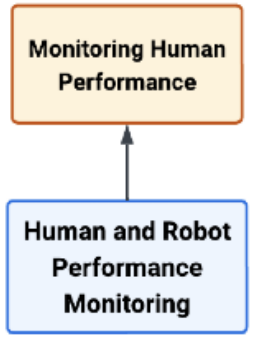
Scope

- Requirements specifically relating to aspects of **human–robot teamwork**
- Applicable across domains and autonomy levels
- Focused on *what requirements ensure*, not *how they are implemented*

Assumptions

- Humans and robots are treated as **team members**
- Teamwork involves shared goals, interdependence, and coordination


Robot Mode Change Notification
Robot System Status Provision
Information Sharing and Awareness
Human Teammate Status Provision
Robot Data Availability


Monitoring Human Performance
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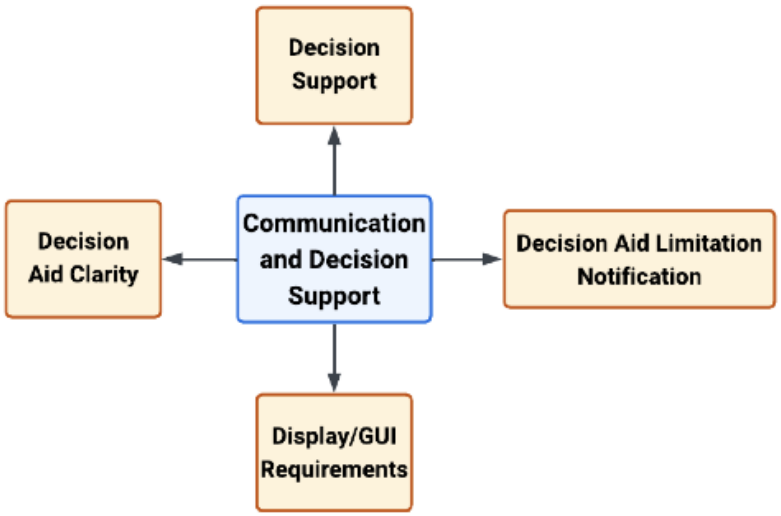
Decision Support
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Decision Aid Limitation Notification
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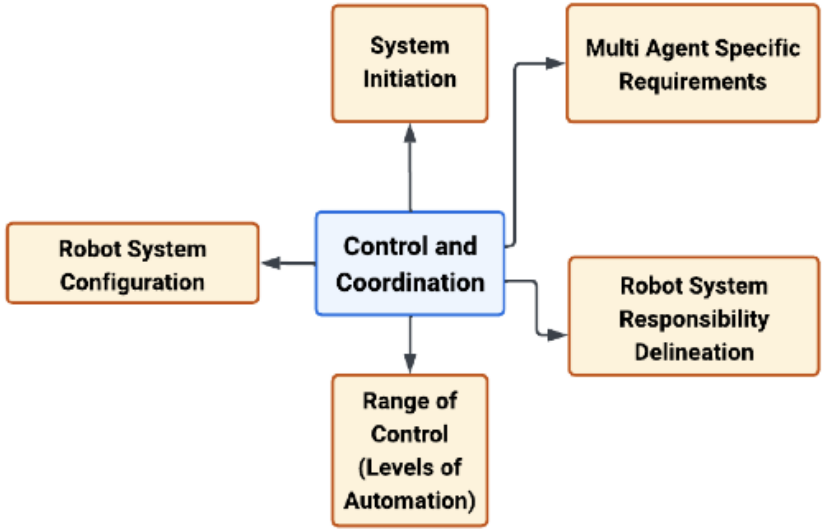
System Initiation
Multi Agent Specific Requirements
Robot System Configuration
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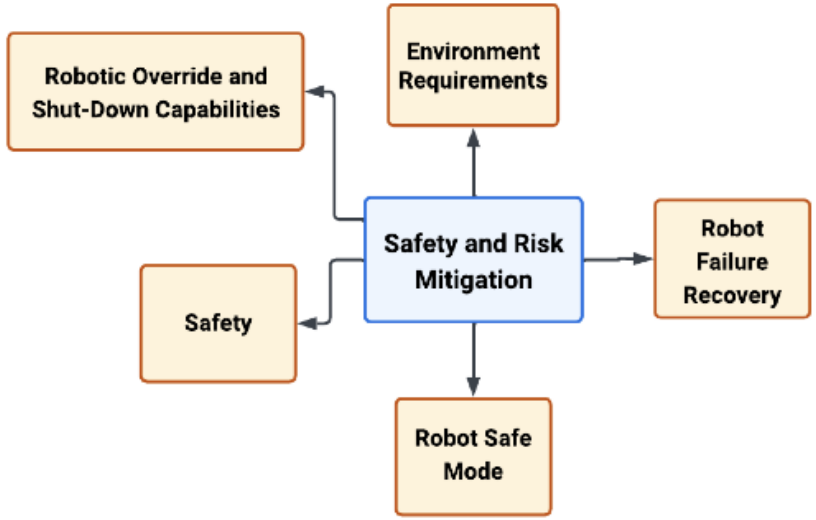
Robot System Responsibility Delineation
Range of Control (Levels of Automation)

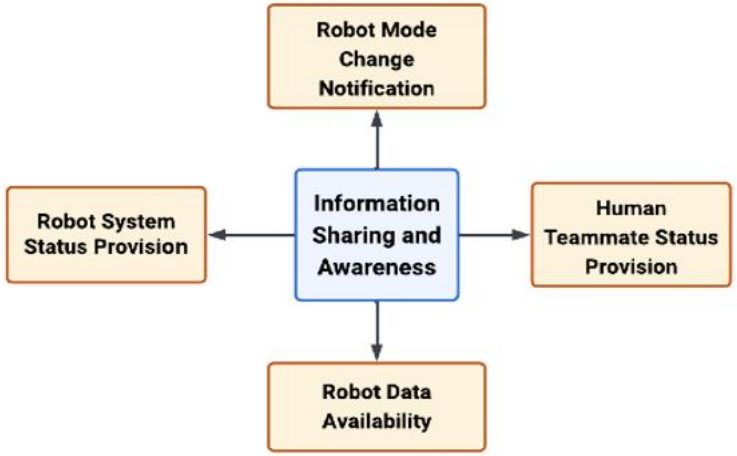
Robot Capabilities/ Characteristics
System Capabilities and Human Control
Human-focused Requirements


Robotic Override and Shut-Down Capabilities
Environment Requirements
Safety and Risk Mitigation
Robot Failure Recovery
Safety
Robot Safe Mode

1. Information Sharing and Awareness

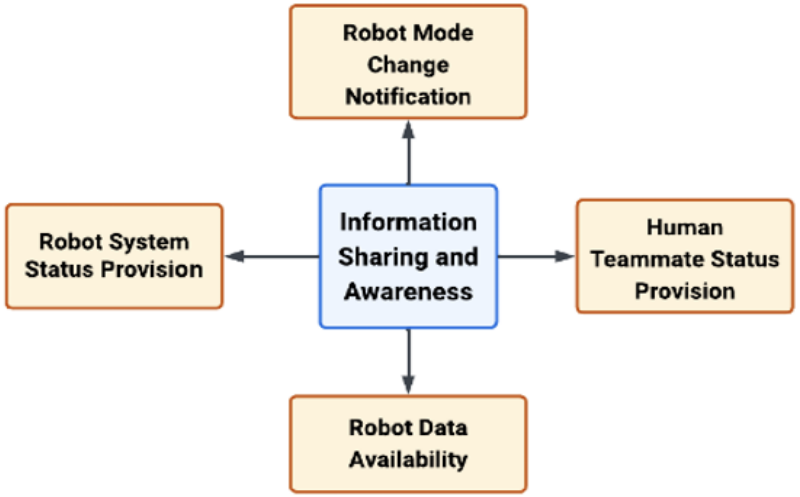
Rationale: This category focuses on the availability and accessibility of critical data for both human and robotic teammates. It emphasizes maintaining situational awareness and ensuring that necessary information is provided to both parties to enable effective collaboration.

- What is exchanged and why:
  - Type of information: Raw data and status updates between human(s) and robot(s)
  - Purpose: Ensuring that both human(s) and robot(s) have the necessary information to understand the system's state and operate effectively



1. Information Sharing and Awareness

Example: The robot provides telemetry, automation status, and mode changes to ensure the human knows what's happening.

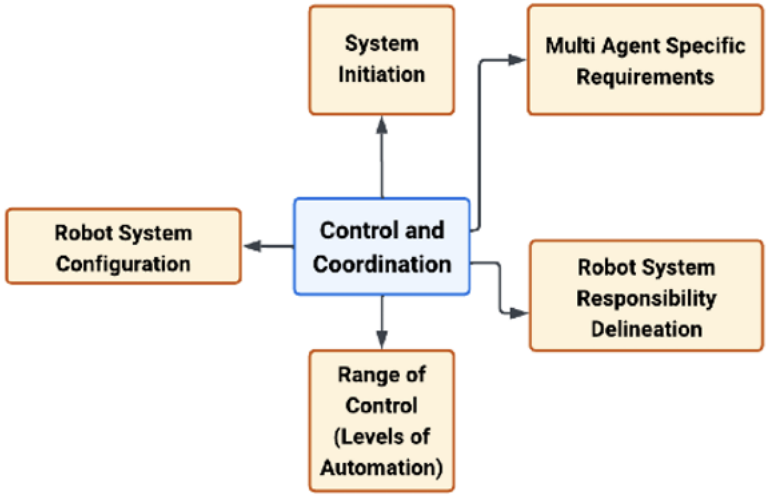


- Is "awareness without action" a meaningful requirement distinction in practice?
- Do you think anything is missing from this category?

2. Control and Coordination

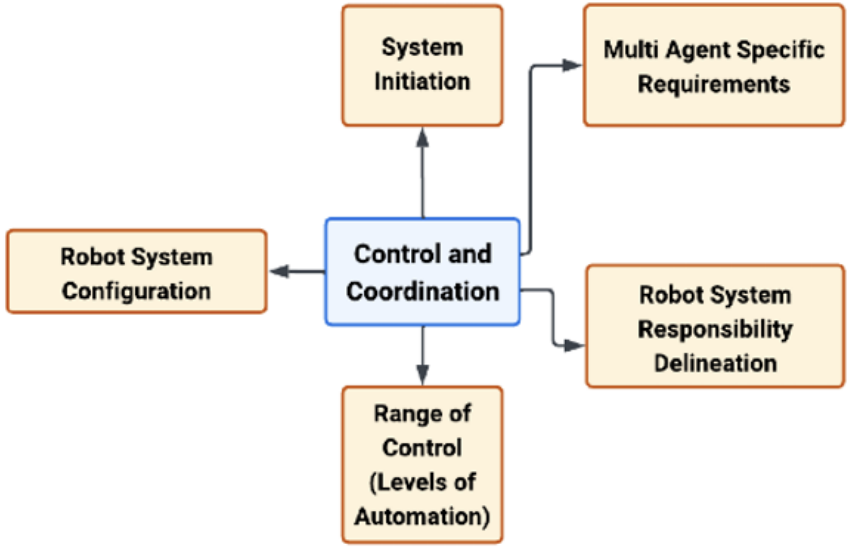
Rationale: This category encapsulates the ability of human operators to configure, control, and delineate responsibilities within the human-robot team. It ensures that the robot is used and performs in a way that aligns with human intent while maintaining safety and operational efficiency.

- What is exchanged and why:
  - Type of information: Commands, permissions, and role delineation between human and robot
  - Purpose: Enabling the human to define, adjust, and/or authorize the robot's automated/autonomous actions while ensuring that the system operates within the defined framework



2. Control and Coordination

Example: The human sets robotic automation levels, defines responsibility boundaries, and/or authorizes system restarts after failures.

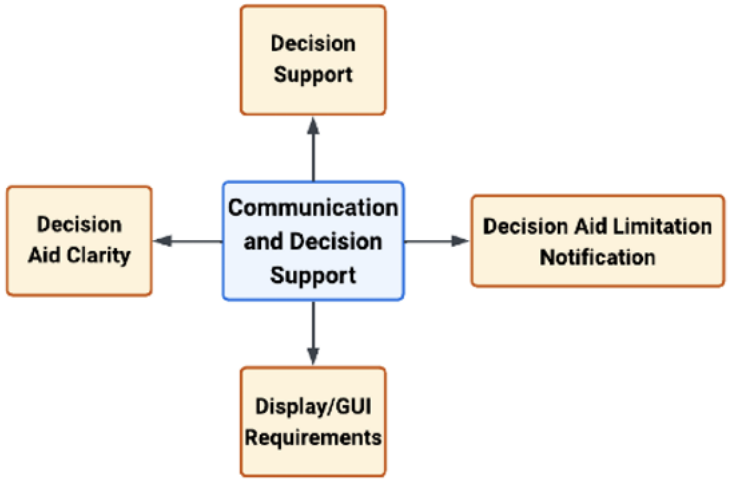


- What are the important parts of control handover that should be captured in requirements?
- Do you think anything is missing from this category?

3. Communication and Decision Support

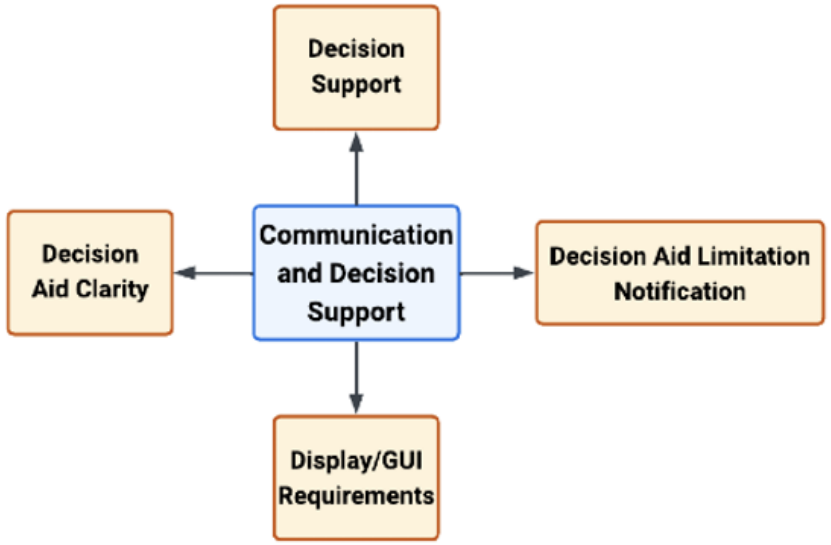
Rationale: Requirements in this category centre on the *provision and clarity* of information. This category highlights the role of communication in facilitating decision-making and understanding between human and robot teammates. It includes requirements that are related to decision aids, interface clarity and notifications to support effective human-robot teamwork

- What is exchanged and why:
  - Type of Information: The exchanged information includes processed information, recommendations, and alerts from the robot to the human
  - Purpose: Helping the human to make better, faster, and more informed decisions while maintaining situational awareness and trust in the robot



3. Communication and Decision Support

- ***Example:*** A decision aid provides suggested actions, explains its reasoning, and notifies the human when it lacks enough data to provide recommendations.

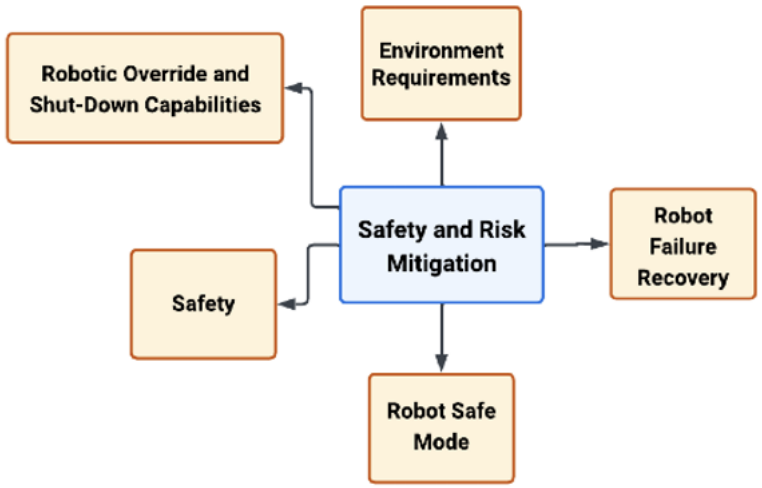


- Are there kinds of teamwork that would be challenging to define in this category?
- Do you think anything is missing from this category?

4. Safety and Risk Mitigation

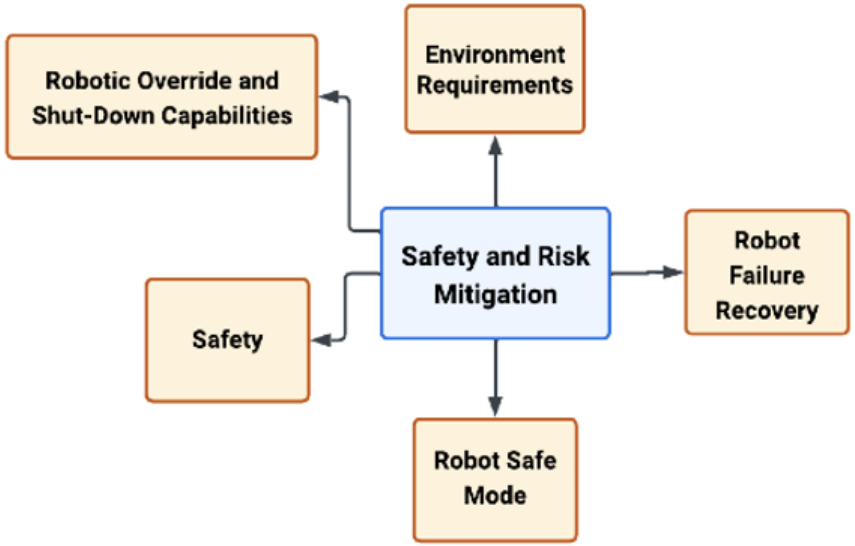
Rationale: This category comprises requirements that ensure safe operation within human-robot collaborative environments. It includes preventative safety measures, emergency responses, and failure recovery mechanisms to protect both human operators and robotic systems

- What is exchanged and why:
  - Type of information: Safety mechanisms, failure responses, and environmental safeguards between the human, robot, and workspace
  - Purpose: Ensuring that the human-robot team operates within a safe framework, minimizing the risks to both human and robotic agents



4. Safety and Risk Mitigation

Example: If the robot fails, it enters a safe mode and/or provides alerts to the human. Safety zones ensure that humans and robots operate without interference.

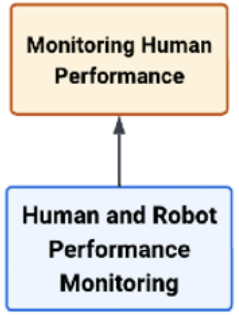


- Have you seen this before?
- Do you think this encapsulates the safety of the interaction, not necessarily the safety of the system?

5. Human and Robot Performance Monitoring

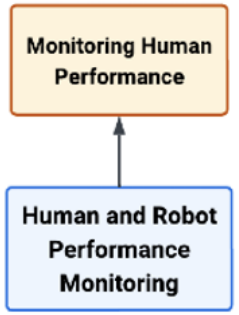
Rationale: This category covers the continuous assessment of both human and robot performance to maintain situational awareness and ensure optimal teamwork. It includes monitoring human behavior as well as robotic system performance

- What is exchanged and why:
  - Type of information: Observations of human and robot behavior, performance metrics, and real-time monitoring data
  - Purpose: Ensuring that both human and robotic agents are performing optimally and adjusting operations based on observed performance



5. Human and Robot Performance Monitoring

Example: The system detects operator fatigue or degraded robot performance and adjusts operations accordingly.



- Does this capture both human and robot performance concerns?
- Have you seen requirements like this before?

6. System Capabilities and Human Control

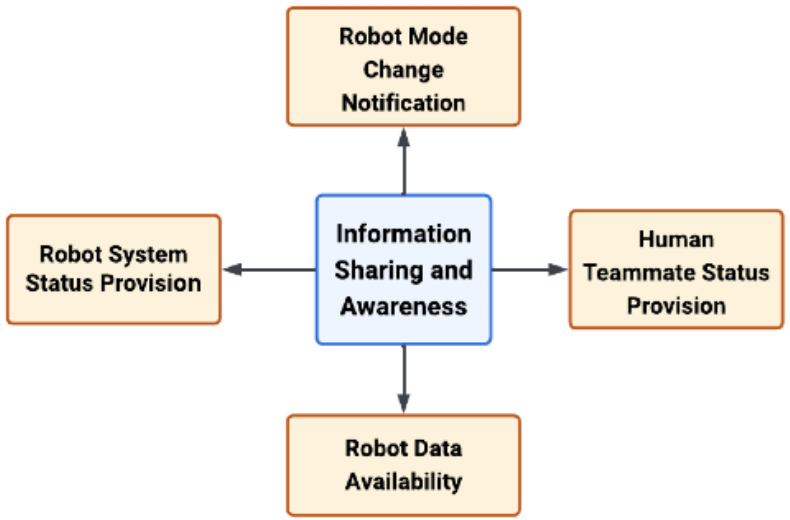
Rationale: This category defines the fundamental capabilities of both robotic and human teammates, ensuring that robots are designed with characteristics that support effective collaboration, while also guaranteeing that humans maintain ultimate authority over the system

- What is exchanged and why:
  - Type of information: System design considerations, human-focused requirements, and override mechanisms
  - Purpose: Establishing the fundamental capabilities of both human and robot teammates while ensuring that the human retains ultimate authority

6. System Capabilities and Human Control

Example: The robot's designed capabilities (e.g., sensor accuracy, movement speed) determine how well it can collaborate. Meanwhile, the human retains the ability to override or shut down the robot at any time.

- Is it useful to separate foundational system capabilities from runtime coordination?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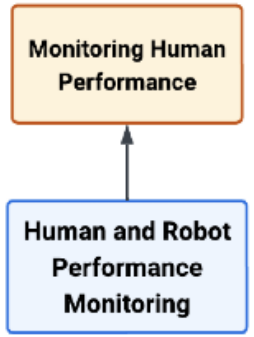
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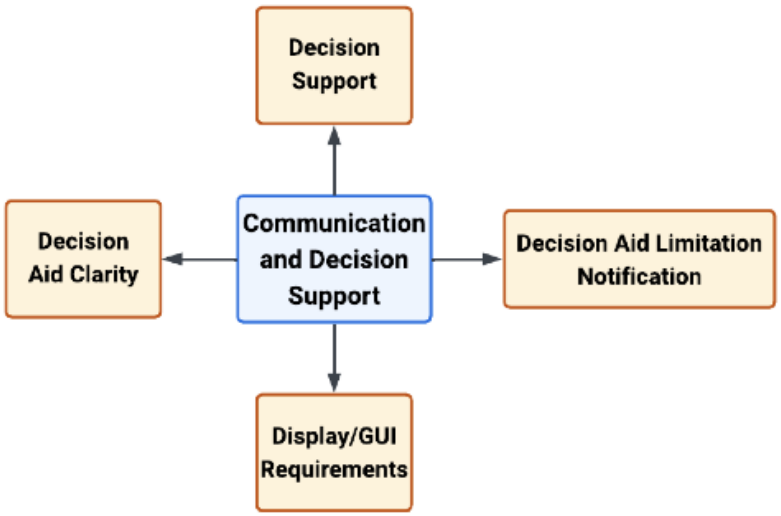
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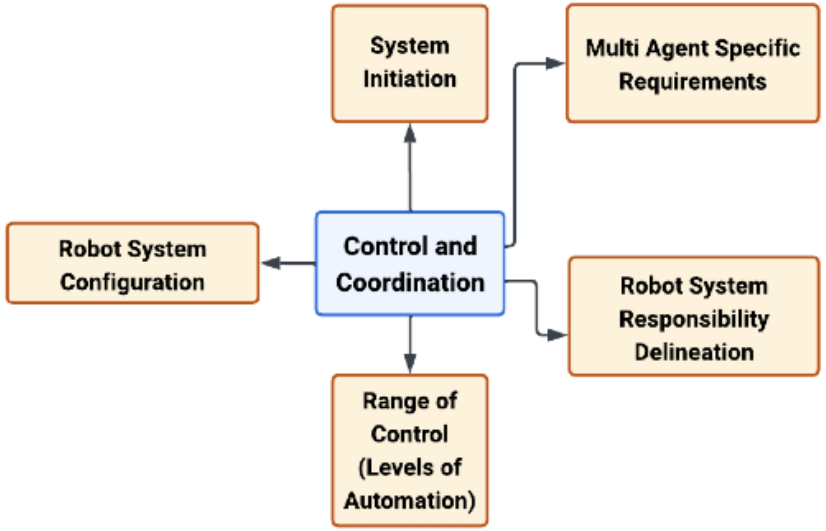
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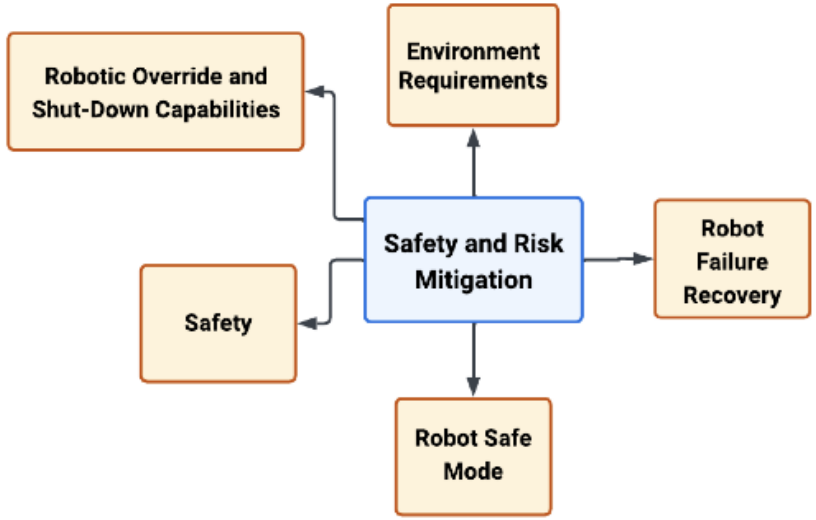
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5. Which categories would be hardest to apply consistently?